

## LIST OF PUBLICATIONS BY ANTON SELITSKIY

### Books

1. A. Selitskii, *Introduction to the Semigroup Theory of Linear Operators and its Application to the Mathematical Physics Problems*. Moscow: MAX Press, 2010. ISBN 978-5-317-03453-5 (in Russian).

### Refereed Journal Publications

Journal rank is shown in brackets where available.

2. G. Zhu, F. Jiang, J. Darefsky, A. Selitskiy, and Zh. Duan, Music Source Separation with Generative Flow // *IEEE Signal Processing Letters*, 29 (2022), 2288–2292.  
DOI: [10.1109/LSP.2022.3219355](https://doi.org/10.1109/LSP.2022.3219355) [Q1 (A\*)]
3. A. Selitskii, On the solvability of parabolic functional differential equations in Banach spaces II // *Eurasian Math. J.*, 11:2 (2020), 86–92. DOI: [10.32523/2077-9879-2020-11-2-86-92](https://doi.org/10.32523/2077-9879-2020-11-2-86-92) [Q2 (B)]
4. A. Selitskii, On the solvability of parabolic functional differential equations in Banach spaces // *Eurasian Math. J.*, 7:4 (2016), 85–91. [Q2 (B)]
5. A. Selitskii,  $L_p$ -solvability of parabolic problems with operator satisfying the Kato conjecture // *Diff. Urav.*, 2015. V. 51, N 6, P. 764–770. English transl. in: *Differential Equations*, 2015. V. 51, N 6, P. 776–782. DOI: [10.1134/S0012266115060087](https://doi.org/10.1134/S0012266115060087) [Q2 (B)]
6. A. Selitskii, Investigation of a mathematical model of nonlinear optical systems with 2-D feedback // *Matem. Mod.* 2015. V. 27, No. 7, P. 117–121 (in Russian).
7. A. Selitskii, Weak and strong solvability in  $L_p$ -spaces of the second boundary-value problem for parabolic differential-difference equation // *Contemporary Analysis and Applied Mathematics*. 2014. V. 2, N 2. P. 226–231. DOI: [10.18532/CAAM.65351](https://doi.org/10.18532/CAAM.65351)
8. A. Selitskii, Space of initial data for the second boundary-value problem for a parabolic differential-difference equation in Lipschitz domains // *Mat. Zametki*. 2013. V. 94, No. 3. P. 477–480. English transl. in: *Mathematical Notes*. 2013. V. 94, N 3. P. 135–138.  
DOI: [10.1134/S0001434613090150](https://doi.org/10.1134/S0001434613090150) [Q2 (B)]
9. A. Selitskii, The space of initial data for the Robin boundary-value problem for parabolic differential-difference equations // *Contemporary Analysis and Applied Mathematics*. 2013. V. 1, N 2. P. 91–97.
10. A. Selitskii, The space of initial data for the second boundary-value problem for parabolic differential-difference equation // *Contemporary Analysis and Applied Mathematics*. 2013. V. 1, N 1. P. 34–41.
11. M. Agranovich and A. Selitskii, Fractional powers of operators corresponding to coercive problems in Lipschitz domains // *Funktsional. Anal. i Prilozhen.* 2013. V. 47. Iss. 2. P. 2–17. English transl. in: *Functional Analysis and Its Applications*. 2013. V. 47, N 2. P. 83–95.  
DOI: [10.1007/S10688-013-0013-0](https://doi.org/10.1007/S10688-013-0013-0) [Q2 (B); top 10% cited in Mathematics 2013, Clarivate Analytics]

12. A. Selitskii, The space of initial data for the second boundary-value problem for parabolic differential-difference equation in 2-D case // *Advancement and Development in Mathematical Sciences*. 2012. V. 2, N 1-2. P. 1–17.
13. A. Selitskii, The modeling of some optical systems on the base of parabolic differential-difference equation // *Matem. Mod.* 2012. V. 24, No. 12, P. 38–42 (in Russian).
14. A. Selitskii, The space of initial data of the 3d boundary-value problem for a parabolic differential-difference equation in the one-dimensional case // *Mat. Zametki*. 2012. V. 92, No. 4. P. 636–640. English transl. in: *Mathematical Notes*. 2012. V. 92, N 4. P. 580–584.  
DOI: [10.1134/S0001434612090313](https://doi.org/10.1134/S0001434612090313) [Q2 (B)]
15. A. Selitskii, The space of initial data of the second boundary-value problem for parabolic differential-difference equation // *Belgorod State University Scientific Bulletin. Mathematics & Physics*. 2011. No. 23(118). Iss. 25. P. 102–111 (in Russian).
16. A. Selitskii and A. Skubachevskii, The second boundary-value problem for parabolic differential-difference equations // *Tr. Semim. im. I. G. Petrovskogo*, 2007. Iss. 26. P. 323–346. English transl. in: *Journal of Mathematical Sciences*, 2007. V. 143, N 4. P. 3386–3400.  
DOI: [10.1007/S10958-007-0216-0](https://doi.org/10.1007/S10958-007-0216-0)
17. A. Selitskii, The third boundary value problem for parabolic differential-difference equation in one-dimensional case // *Functional Differential Equations*, 2007. V. 14. P. 373–395. [Q2 (B)]
18. A. Selitskii and A. Skubachevskii, Second boundary-value problem for parabolic differential-difference equations // *Uspekhi Mat. Nauk*, 2007. V. 62. Iss. 1. P. 207–208. English transl. in: *Russian Mathematical Surveys*, 2007. V. 62, No. 1. P. 191–192.  
DOI: [10.1070/RM2007V062N01ABEH004391](https://doi.org/10.1070/RM2007V062N01ABEH004391) [Q2 (B)]
19. A. Selitskii, The third boundary-value problem for parabolic differential-difference equations // *Contemporary Mathematics. Fundamental Directions*, 2007. V. 21. P. 114–132. English transl. in: *Journal of Mathematical Sciences*, 2008. V. 153, N 5. P. 591–611.  
DOI: [10.1007/S10958-008-9138-8](https://doi.org/10.1007/S10958-008-9138-8)

## Refereed Conference Publications

20. A. Selitskiy and D. Millard, One-Step Voice Conversion by Learning kNN Transport in WavLM Space // [Under review]
21. A. Selitskiy and D. Millard, Stability of semi-dual neural optimal transport // [Under review]
22. A. Selitskiy, A. Shahriyar, and J. Prakasan, Discrete optimal transport is a strong audio adversarial attack // IEEE/IAPR International Joint Conference on Biometrics (IJCB), Rome, Italy, September 1–4, 2026. [In press] [B]
23. A. Selitskiy and M. Kocharekar, Discrete optimal transport and voice conversion // *2026 11th International Conference on Machine Learning Technologies (ICMLT)*, Berlin, Germany, May 20–22, 2026. P. 318–322. DOI: [10.1109/ICMLT69916.2026.11688511](https://doi.org/10.1109/ICMLT69916.2026.11688511)
24. D. Millard and A. Selitskiy, Deep Ritz method for elliptic differential-difference equations // AMS Spring Eastern Sectional Meeting, April 5–6, 2025, Hartford, CT.
25. A. Selitskii, On the smoothness of solutions of the second boundary-value problem for parabolic differential difference equation // Proceedings of the III International Conference: Mathematical

Ideas by P.L. Chebyshev and Their Applications to the Problems of Science. Obninsk, 2008. P. 144–150.

## Conferences

26. A. Selitskii, Deep Ritz method for parabolic differential-difference equations // 2024 New York New Jersey Pennsylvania Section of SIAM Annual Meeting (poster).
27. G. Zhu, F. Jiang, J. Darefsky, A. Selitskiy, and Zh. Duan, Music Source Separation with Generative Flow // North East Music Information Special Interest Group 2022 (New Jersey) (poster).
28. A. Selitskii, On a new class of operators satisfying the Kato conjecture // The International Conference Mathematical and Computational Modelling in Science and Technology. Izmir, Turkey (August 02–07, 2015). Abstract Book. Izmir: Izmir University, 2015. P. 162–163. (20-minute communication)
29. A. Selitskii, Investigation of a mathematical model of nonlinear optical systems with 2-D feedback // Modern Problems of Applied Mathematics and Informatics. Dubna, Russia, August, 2014. P. 187–189. (15-minute communication)
30. A. Selitskii, Maximal regularity of parabolic problems with operator satisfying the Kato conjecture // The Seventh International Conference on Differential and Functional Differential Equations. Abstracts. Moscow, Russia, August 15–29, 2014. Moscow: Steklov Mathematical Institute, Lomonosov Moscow State University, Moscow Mathematical Society, 2014. P. 70. (15-minute communication)
31. A. Selitskii, The solvability in  $L_p$  spaces of parabolic equations with operators satisfying Kato's conjecture // International Conference "Spectral Theory and Differential Equations", dedicated to the Century of B.M. Levitan. Moscow, Russia, June, 2014. P. 117–118. (15-minute communication)
32. A. Selitskii, Maximal regularity of the 2-d mixed problem for parabolic differential-difference equations // Program of IWOTA 2014. Amsterdam. VU University. P. 94–95. (15-minute communication)
33. M. Agranovich and A. Selitskii, Strongly elliptic systems in Lipschitz domains and the Kato conjecture // International conference KROMSH-2013. Crimea, September, 2013. (45-minute lecture)
34. A. Selitskii and A. Skubachevskii, The space of initial data for parabolic differential-difference equations // Abstracts. 9th International ISAAC Congress. August 5–9, 2013 in Krakow, Poland. Gliwice: PKJS, 2013. P. 185. (15-minute communication)
35. M. Agranovich and A. Selitskii, Elementary approach to fractional powers of strongly elliptic operators in Lipschitz Domains // Abstracts. 9th International ISAAC Congress. August 5–9, 2013 in Krakow, Poland. Gliwice: PKJS, 2013. P. 16. (15-minute communication)
36. M. Agranovich and A. Selitskii, Elementary approach to fractional powers of strongly elliptic operators in Lipschitz Domains // 5-th St. Petersburg Conference in Spectral Theory dedicated to the memory of M.Sh. Birman. July 2–6, 2013. P. 8. (30-minute communication)

37. A. Selitskii and A. Skubachevskii, On the strong solution for the second parabolic problem with differential-difference operator // International conference “Contemporary methods and problems of operator theory and harmonic analysis”. Rostov-na-Donu, Russia, June, 2013. P. 80. (15-minute communication)
38. M. Agranovich and A. Selitskii, Fractional powers of operators corresponding to boundary-value problems in Lipschitz domains // International conference “Contemporary methods and problems of operator theory and harmonic analysis”. Rostov-na-Donu, Russia, June, 2013. P. 47. (45-minute lecture)
39. A. Selitskii, The space of initial data for the third boundary problem for parabolic differential-difference equation in one-dimensional case // III International Conference: Mathematical Ideas by P.L. Chebyshev and Their Applications to the Problems of Science. Obninsk, Russia, May, 2006. P. 106–107. (15-minute communication)
40. A. Selitskii, The third boundary value problem for parabolic differential-difference equation // The XLII Russian Conference on the Problems of Mathematics, Informatics and Chemistry. Moscow, Russia, April, 2006. P. 14. (15-minute communication)
41. A. Selitskii, The second mixed problem for parabolic differential-difference equation // The Fourth International Conference on Differential and Functional Differential Equations. Abstracts. Moscow, Russia, August 14–21, 2005. Moscow: Steklov Mathematical Institute, Lomonosov Moscow State University, Moscow Mathematical Society, 2005. P. 70–71. (15-minute communication)
42. G. Izmailov and A. Selitskii, The solution of oscillation equation for a superconductive circuit // IV International Scientific Conference “The Chkalov Meeting”. Engineering and Physical Problems of Aviation and Space Technology. Egoryevsk, Russia, June, 2002. (15-minute communication)

## Other Publications

43. A. Selitskiy, *Optimal Transport in Audio Signal Processing. Submitted in Partial Fulfillment of the Requirements for the Degree Doctor of Philosophy*. Rochester: ProQuest Dissertations and Theses, 2026. ISBN 978-8-244-85748-1.
44. A. Selitskiy, *The Second and the Third Boundary Value Problems for Parabolic Differential-Difference Equation. Thesis Submission for a Degree of Candidate of Science in Physics and Mathematics*. Moscow, 2007.